

JEE-Main-08-04-2026 (Memory Based)
[EVENING SHIFT]
Physics

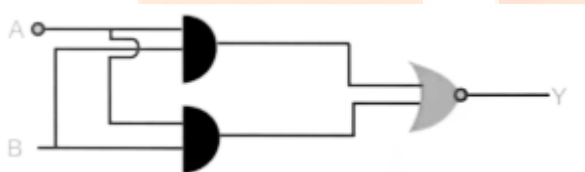
Question: A balloon is moving with speed 10 m/s in upward direction. At height of 75 m a stone is released. Distance travelled by stone in air.

Options:

- (a) 70 m
- (b) 80 m
- (c) 85 m
- (d) 90 m

Answer: (c)

Question: For the logic gate shown in the diagram, find the output Y for the given input A and B.



Options:

- (a) $A \cdot \overline{B}$
- (b) $\overline{A} + \overline{B}$
- (c) $\overline{A + B}$
- (d) $\overline{A} \cdot \overline{B}$

Answer: (b)

Question: Two photons of wavelength λ_1 and λ_2 ($\lambda_1 = 2\lambda_2 = 2\lambda$) use photoelectric effect on a metal surface emitting two photoelectrons of kinetic energies K_1 and K_2 respectively. Find work function of metal ($K_1 = 3K_2 = 3K$)

Options:

- (a) $\frac{6hc}{\lambda}$
- (b) $\frac{5hc}{4\lambda}$
- (c) $\frac{hc}{2\lambda}$

- (d) $\frac{4hc}{3\lambda}$
Answer: (b)

Question: A block attached to a spring has natural frequency f_1 . If the spring is cut in two equal half and only one of the half spring is connected to the block then the

frequency becomes f_2 . Find $\frac{f_2}{f_1}$

Options:

- (a) 2
 (b) $\frac{1}{2}$
 (c) $\sqrt{2}$
 (d) $1/\sqrt{2}$

Answer: (c)

Question: Two projectiles projected from ground with same speed have same range, their time are 5 s and 10 s respectively. Find their range.

Options:

- (a) 125 m
 (b) 250 m
 (c) 100 m
 (d) 480 m

Answer: (b)

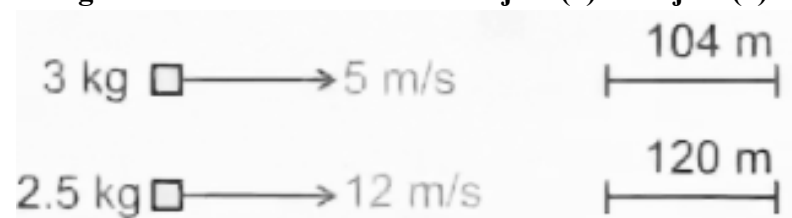
Question: A new unit (a) of length is chosen such that it is equal to the distance travelled by light in vacuum in 1 second. What is the distance d between Venus and Earth in terms of this new unit, if light takes 6 minutes 40 seconds to cover the distance?

Options:

- (a) 500 unit
 (b) 300 unit
 (c) $1/200$ unit
 (d) 400 unit

Answer: (d)

Question: Two objects (1) and (2) travel 104 m and 120 m respectively in the 50th second, with initial speeds of 5 m/s and 12 m/s respectively. Their masses are 3 kg and 2.5 kg. the ratio of momentum of object (1) to object (2) at $t = 50$ sec is nearly:



Options:

- (a) 2
 (b) 4
 (c) 1
 (d) 3

Answer: (c)

Question: Dipole moment of a system, with q_1 and q_2 at

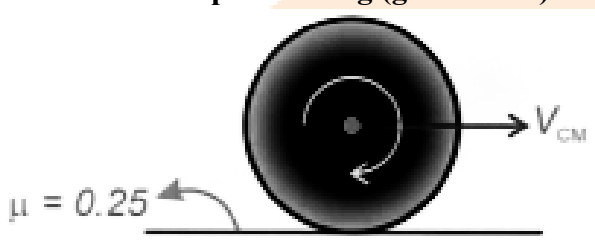
$\vec{r}_1$ and $\vec{r}_2$ as μC , $-9 \mu C$ and $\vec{r}_1 = 2\hat{i} + 3\hat{j} + 3\hat{k}$, $\vec{r}_2 = \hat{i} + \hat{j} + \hat{k}$ respectively is of magnitude

Options:

- (a) $5\sqrt{2} \mu C m$
- (b) $10 \mu C m$
- (c) $3 \mu C m$
- (d) $12 \mu C m$

Answer: (c)

Question: A solid cylinder has $V_{cm} = 9 \text{ m/s}$ & $\omega = \frac{v_{cm}}{4R}$ as shown. Find time after which it will start pure rolling ($g = 10 \text{ m/s}^2$)



Options:

- (a) 1 s
- (b) 0.9 s
- (c) 2 s
- (d) 1.3 s

Answer: (b)

Question: Initial pressure and volume of monoatomic gas is P & V. It is expanded adiabatically to 27 times its initial volume Find magnitude of change in internal energy.

Options:

- (a) $\frac{3}{2} PV$
- (b) PV
- (c) $\frac{4}{3} PV$
- (d) $\frac{PV}{2}$

Answer: (c)

Question: Magnetic field is represented in terms of μ_0 , ϵ_0 and electric field E as $\mu_0^x \epsilon_0^y E^z$ then (x + y + z) is

Options:

- (a) 1.5
- (b) 0.5
- (c) 2
- (d) 2.5

Answer: (c)

Question: A solenoid is having 10 turns, length of 25 mm and of cross section area of 5 cm^2 . The current in solenoid changes from 2A to 4A in 3.14 s then average EMF induced is

Options:

- (a) 0.9 mV
- (b) 0.35 mV
- (c) $0.16 \mu\text{V}$
- (d) $0.8 \mu\text{V}$

Answer: (c)

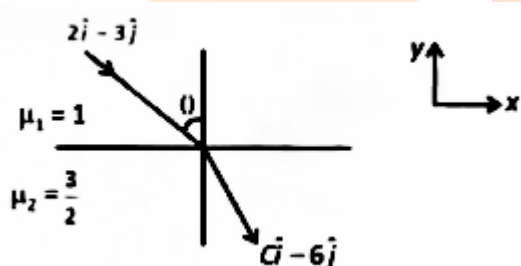
Question: A car is moving with speed of 54 km/h takes a turn of radius 20 m. Angle made by bob from car ceiling with vertical is

Options:

- (a) $\tan^{-1}(1.125)$
- (b) $\tan^{-1}(0.05)$
- (c) $\tan^{-1}(0.09)$
- (d) $\tan^{-1}(0.6)$

Answer: (c)

Question: The direction (vector) of incident and refracted light are shown in the diagram. Find the value of C.



Options:

- (a) $\frac{24}{\sqrt{101}}$
- (b) $\frac{24}{7}$
- (c) 3
- (d) $\frac{25}{\sqrt{101}}$

Answer: (a)

Question: A drop is falling with terminal velocity v_1 in a gas. The drop splits into 64 identical droplets. If one small droplet falls with terminal velocity v_2 , find the ratio $\frac{v_1}{v_2}$

Answer: (16)

JEE-Main-08-04-2026 (Memory Based)
[EVENING SHIFT]
Chemistry

Question: Increasing order of bond length from the following

Options:

- (a) O_2^+ , O_2 , O_2^{2-} , O_2^-
- (b) O_2^- , O_2^{2-} , O_2^+ , O_2
- (c) O_2^+ , O_2 , O_2^- , O_2^{2-}
- (d) O_2^{2-} , O_2^- , O_2 , O_2^+

Answer: (d)

Solution: (d) O_2^{2-} , O_2^- , O_2 , O_2^+

Question: Given

$\lambda_1 = 3000 \text{ \AA}$ and $\lambda_2 = 6000 \text{ \AA}$ then find the ratio of E_1 and E_2

Options:

Answer: (2)

Solution:

Question: If $E_{Fe^{2+}/Fe}^0$ is xV and $E_{Fe^{3+}/Fe}^0$ is yV .

Then calculate $E_{Fe^{2+}/Fe^{3+}}^0$ in V .

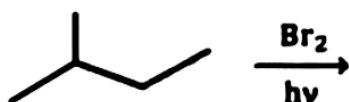
Options:

- (a) $(2x - 3y)$
- (b) $(2x + 3y)$
- (c) $(x - y)$
- (d) $(x + y)$

Answer: (a)

Solution: (a) $(2x - 3y)$

Question: The number of monobrominated products in the following reaction is (including stereoisomers)



Options:

Answer: (6)

Solution:

Question: Which of the following given violet color complex with natural FeCl_3 solution?

Options:

- (a) Threonine
- (b) Serine
- (c) Tyrosine
- (d) Cysteine

Answer: (c)

Solution: (c) Tyrosine

Question: Consider the following



What is K_3 for $\frac{1}{2} \text{X}_2 + \frac{1}{2} \text{Y}_2 + \frac{1}{2} \text{Z}_2 \rightleftharpoons \text{XYZ}$

Options:

- (a) 5×10^{-4}
- (b) 2.5×10^{-5}
- (c) 1×10^4
- (d) 2.5×10^{-4}

Answer: (b)

Solution: (b) 2.5×10^{-5}

Question: Two photons of wavelength λ_1 and λ_2 ($\lambda_1 = 2\lambda_2 = 2\lambda$) use photoelectric effect on a metal surface emitting two photoelectrons of kinetic energies K_1 and K_2 respectively. Find work function of metal ($K_1 = 3K_2 = 3K$)

Options:

- (a) $\frac{6hc}{\lambda}$
- (b) $\frac{5hc}{4\lambda}$
- (c) $\frac{hc}{2\lambda}$
- (d) $\frac{4hc}{3\lambda}$

Answer: (b)

Solution: (b) $\frac{5hc}{4\lambda}$

Question: Match the Lists

List-I (n = 2)	List-II (IE in kJ/mol)
A) ns^2	1) 1402
B) ns^2np^1	2) 99
C) ns^2np^3	3) 2080
D) $ns^2 np^6$	4) 801

Options:

- (a) A→2, B→4, C→3, D→1
- (b) A→4, B→2, C→1, D→3
- (c) A→2, B→4, C→1, D→3

(d) A→4, B→2, C→3, D→1

Answer: (c)

Solution: (c) A→2, B→4, C→1, D→3

Question: Bromine trifluoride auto ionic to form BrF_2^+ and BrF_4^- . The shapes of Cation and anion are respectively

Options:

- (a) Linear, square planar
- (b) Bent, See-Saw
- (c) Bent, Square planar
- (d) Linear, tetrahedral

Answer: (c)

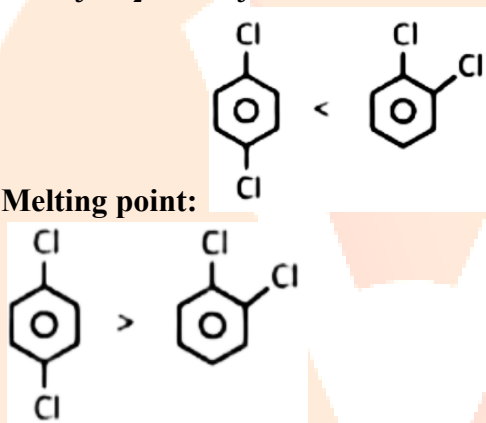
Solution: (c) Bent, Square planar

Question: Given below are two statements.

Statement-I Boiling point follows the order

$\text{CH}_3\text{CH}_2\text{CH}_2\text{I} > \text{CH}_3\text{CH}_2\text{I} > \text{CH}_3\text{I}$.

Statement-II: Melting point:



Boiling point:

Choose the correct statements.

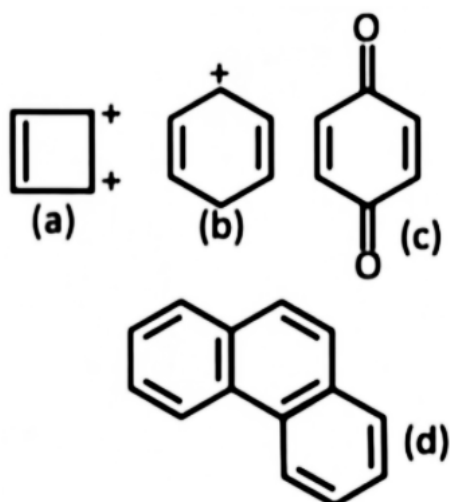
Options:

- (a) Both Statement-I and Statement-II are correct
- (b) Statement-I is correct but Statement-II is incorrect
- (c) Statement-I is incorrect but statement-II is correct
- (d) Both Statement-I and Statement-II are incorrect

Answer: (B)

Solution: (b) Statement-I is correct but Statement-II is incorrect

Question: The number of aromatic compounds from the following is



Options:

Answer: (2)

Solution:

Question: Given below are two statements

Statement-I: Given $E_a = 12.6 \text{ K cal}$; then at room temperature, the rate constant double for a rise of 10°C

Statement-II: In a first order reaction

Choose the correct option



Options:

- (a) Both Statement-I and Statement-II are true
- (b) Statement-I is false but statement-II is true
- (c) Statement-I is true but statement-II is false
- (d) Both Statement-I and Statement-II are false

Answer: (c)

Solution: (c) Statement-I is true but statement-II is false

Question: Given below are two statements

Statement-I: 30% (w/w) Solution of CCl_4 in Benzene has mole fraction of CCl_4 to be 0.033

Statement-II A mixture of CCl_4 and ethanol shows positive deviation

In light of above statements, choose the correct option

Options:

- (a) Both Statement-I and Statement-II are correct
- (b) Statement-I is correct statement-II is incorrect
- (c) Statement-I is incorrect statement-II is correct
- (d) Both Statement-I and Statement-II are incorrect

Answer: (c)

Solution: (c) Statement-I is incorrect statement-II is correct

Question: Match the following.

List-I (Mass of molecules)	List-II (Number of molecules)
A) 1.8 mg H ₂ O	i) $0.5 \times 10^{-4} N_A$
B) 1.8 mg carbon	ii) $1.5 \times 10^{-4} N_A$
C) 4.9 mg H ₂ SO ₄	iii) $2 \times 10^{-4} N_A$
D) 11.7 mg NaCl	iv) $1 \times 10^{-4} N_A$

Options:

(a) P→III, Q→IV, R→I, S→II

(b) P→VI, Q→II, R→I, S→III

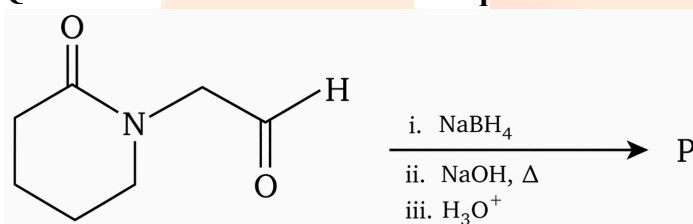
(c) P→I, Q→II, R→III, S→IV

(d) P→II, Q→III, R→I, S→IV

Answer: (b)

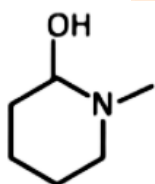
Solution: (b) P→VI, Q→II, R→I, S→III

Question: Consider the reaction sequence:



Identify P

Options:



(a)

(b) $\text{OHC}-(\text{CH}_2)_4-\text{NH}-\text{CH}_2-\text{OH}$

(c) $\text{HO}_2\text{C}-(\text{CH}_2)_4-\text{NH}-\underset{\text{I}}{\text{CH}}-\text{CH}_2-\text{OH}$

(d) $\text{OHC}-(\text{CH}_2)_4-\text{NH}-\underset{\text{I}}{\text{CH}_2}-\text{CH}_2-\text{OH}$

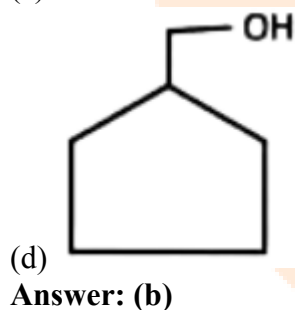
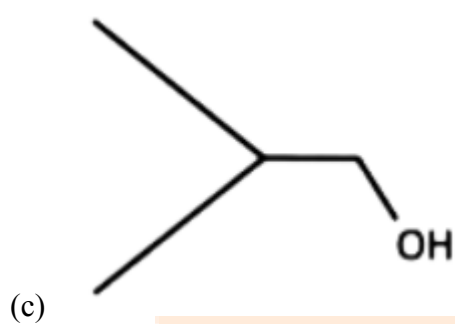
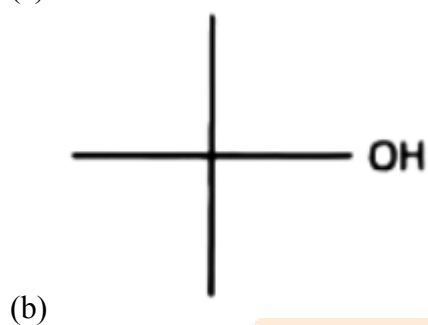
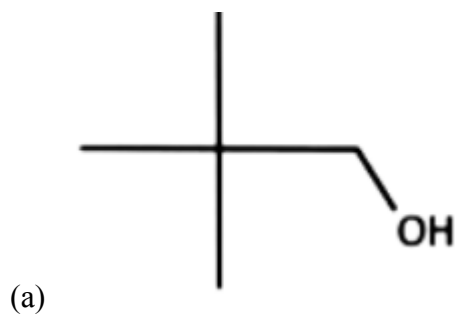
(d)

Answer: (c)

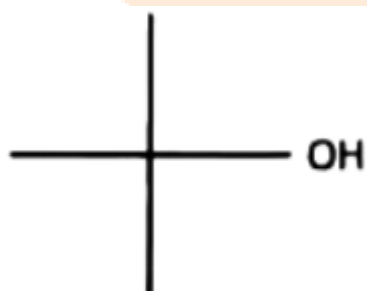
Solution: (c) $\text{HO}_2\text{C}-(\text{CH}_2)_4-\text{NH}-\underset{\text{I}}{\text{CH}}-\text{CH}_2-\text{OH}$

Question:

Options:



Answer: (b)



Solution: (b)

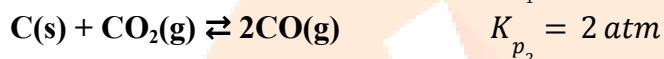
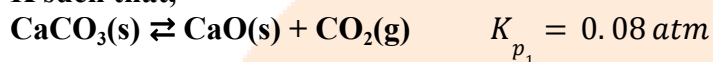
Question: Statement I: Reducing character decreases from NH_3 to BiH_3 .
 Statement II: Tendency to donate a lone pair of e^- decreases from NH_3 to BiH_3 .
 Statement III: Stability of hydrides decreases from NH_3 to BiH_3 .
 Statement IV: The H–E–H bond angle of NH_3 is greater than SbH_3 .
 Which of the following is correct statements?

Options:

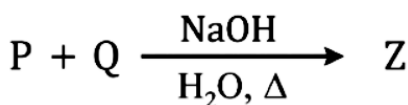
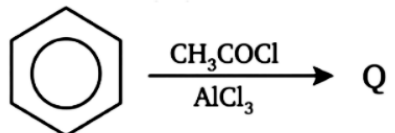
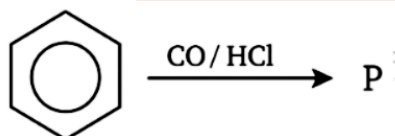
- (a) S_1 , S_2 and S_3 only
 (b) S_2 and S_4 only
 (c) S_2 , S_3 and S_4 only
 (d) S_1 and S_3 only

Answer: (c)**Solution:** (c) S_2 , S_3 and S_4 only**Question:** Number of paramagnetic complexes among the following is**Options:****Answer: (4)****Solution:**

Question: Solid C, CaO and CaCO_3 are mixed and allowed to attain equilibrium at 'T' K such that,



Partial pressure of CO at equilibrium is $x \times 10^{-1}$ atm. X is ____.

Options:**Answer: (4)****Solution:****Question:** Consider the following reaction

The number of π bonds in (Z) is

Options:

- (a) 8
 (b) 10
 (c) 12
 (d) 5

Answer: (a)**Solution:** (a) 8

JEE-Main-08-04-2026 (Memory Based)
[EVENING SHIFT]
Maths

Question: If $\alpha = 3 + 4 + 8 + 9 + 13 + \dots$ upto 40 terms and $\tan(\beta)^{\frac{\alpha}{1020}}$ is the root of the equation $x^2 - x - 2 = 0$. Then, the value of $\sin^2\beta + 3\cos^2\beta$ is

Options:

- (a) $\frac{5}{3}$
- (b) 2
- (c) 1
- (d) $\frac{1}{2}$

Answer: (d)

Solution:

Step 1: Find the value of α

The series for α is: $3 + 4 + 8 + 9 + 13 + 14 + \dots$ up to 40 terms.
 We can group these into 20 pairs:

- 1st pair: $(3 + 4) = 7$
- 2nd pair: $(8 + 9) = 17$
- 3rd pair: $(13 + 14) = 27$

These pair-sums form an Arithmetic Progression (AP) where the first term $a = 7$, the common difference $d = 10$, and the number of terms $n = 20$.

Using the sum formula $S_n = \frac{n}{2} [2a + (n - 1)d]$:

$$\alpha = \frac{20}{2} [2(7) + (20 - 1)10]$$

$$\alpha = 10[14 + 190] = 10 \times 204 = 2040$$

Step 2: Solve the quadratic equation

The equation is $x^2 - x - 2 = 0$. Factorizing it gives:

$$(x - 2)(x + 1) = 0$$

So, the roots are $x = 2$ and $x = -1$.

Step 3: Find $\tan(\beta)$

The problem states that $\tan(\beta)^{\frac{\alpha}{1020}}$ is a root.

Substituting $\alpha = 2040$:

$$\tan(\beta) \cdot \frac{2040}{1020} = \text{root} \implies 2 \tan(\beta) = \text{root}$$

- If the root is 2: $2 \tan(\beta) = 2 \implies \tan(\beta) = 1$
- If the root is -1: $2 \tan(\beta) = -1 \implies \tan(\beta) = -\frac{1}{2}$

Step 4: Calculate the final expression

We need the value of $\sin^2 \beta + 3\cos^2 \beta$. We can rewrite this as:

$$(\sin^2 \beta + \cos^2 \beta) + 2\cos^2 \beta = 1 + 2\cos^2 \beta$$

Using the identity $\cos^2 \beta = \frac{1}{1 + \tan^2 \beta}$:

Using Case 1 ($\tan \beta = 1$):

$$\cos^2 \beta = \frac{1}{1 + 1^2} = \frac{1}{2}$$

$$\text{Value} = 1 + 2\left(\frac{1}{2}\right) = 1 + 1 = 2$$

Question: If $\alpha = 3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3\cos^{-1}\left(\frac{9}{16}\right)$, consider statements:

Statement 1 : $\cos(\alpha + \beta) > 0$

Statement 2 : $\cos \alpha < 0$

Then which of the following is true?

Options:

- (a) Statement 1 and 2 are correct
- (b) Only Statement 1 is correct
- (c) Only Statement 2 is correct
- (d) None of these Statement are correct

Answer: (c)

Solution:

1. Analyze α and Statement 2

We are given $\alpha = 3\sin^{-1}\left(\frac{6}{11}\right)$.

- Let $\theta = \sin^{-1}\left(\frac{6}{11}\right)$. Since $\sin(30^\circ) = 0.5$ and $\frac{6}{11} \approx 0.545$, we know that θ is slightly larger than 30° (specifically, $\theta \approx 33.06^\circ$).
- So, $\alpha = 3\theta \approx 3 \times 33.06^\circ = 99.18^\circ$.
- Since α is between 90° and 180° , it lies in the **second quadrant**.
- In the second quadrant, the cosine function is negative. Therefore, $\cos \alpha < 0$ is **TRUE**.

2. Analyze β and Statement 1

We are given $\beta = 3\cos^{-1}\left(\frac{9}{16}\right)$.

- Let $\phi = \cos^{-1}\left(\frac{9}{16}\right)$. Since $\cos(60^\circ) = 0.5$ and $\frac{9}{16} = 0.5625$, we know that ϕ is slightly smaller than 60° (specifically, $\phi \approx 55.77^\circ$).
- So, $\beta = 3\phi \approx 3 \times 55.77^\circ = 167.31^\circ$.
- Now, let's find the sum $\alpha + \beta$:

$$\alpha + \beta \approx 99.18^\circ + 167.31^\circ = 266.49^\circ$$

- The angle 266.49° lies in the **third quadrant** (which is between 180° and 270°).
- In the third quadrant, the cosine function is negative.
- Statement 1 says $\cos(\alpha + \beta) > 0$, which is **FALSE**.

Question: A person has 3 different bags and 4 different books. The number of ways in which he can put these books in the bags so that no bag is empty, is

Options:

- (a) 36
- (b) 34

(c) 32

(d) 30

Answer: (a)

Solution:

Step 1: Group the books

First, we calculate how many ways we can divide 4 different books into groups of 2, 1, and 1.

Using the formula for grouping:

$$\text{Number of ways to form groups} = \frac{4!}{2! \cdot 1! \cdot 1! \cdot 2!}$$

(Note: We divide by 2! at the end because there are two groups of the same size, i.e., size 1).

$$\frac{24}{2 \times 1 \times 1 \times 2} = 6 \text{ ways to form groups}$$

Step 2: Distribute the groups into bags

Now, we have 3 groups of books that need to be placed into 3 different bags. The number of ways to arrange these 3 groups into 3 bags is:

$$3! = 3 \times 2 \times 1 = 6 \text{ ways}$$

Step 3: Calculate the total

To find the total number of ways, we multiply the grouping ways by the distribution ways:

$$\text{Total ways} = 6 \times 6 = 36$$

Question: If $(x\sqrt{1-x^2})dy - (y\sqrt{1-x^2} - x^2\cos^{-1}x)dx = 0$ and $\lim_{x \rightarrow 1^-} y(x) = 1$, then $y\left(\frac{1}{2}\right)$ is

Options:

- (a) $\frac{\pi^2}{36}$
- (b) $\frac{\pi}{36} + \frac{1}{2}$
- (c) $\frac{\pi^2}{36} + \frac{1}{2}$
- (d) $\frac{\pi}{36}$

Answer: (c)

Solution:

$$x\sqrt{1-x^2} \frac{dy}{dx} = y\sqrt{1-x^2} - x^2 \cos^{-1} x$$

$$\frac{dy}{dx} = \frac{y}{x} - \frac{x \cos^{-1} x}{\sqrt{1-x^2}}$$

$$\frac{dy}{dx} - \frac{1}{x} y = - \frac{x \cos^{-1} x}{\sqrt{1-x^2}}$$

$$\text{I.F.} = e^{\int -\frac{1}{x} dx} = e^{-\ln|x|} = \frac{1}{x}$$

$$\frac{d}{dx} \left(y \cdot \frac{1}{x} \right) = \frac{1}{x} \left(- \frac{x \cos^{-1} x}{\sqrt{1-x^2}} \right)$$

$$\frac{y}{x} = \int - \frac{\cos^{-1} x}{\sqrt{1-x^2}} dx$$

$$\frac{y}{x} = \frac{(\cos^{-1} x)^2}{2} + C \Rightarrow y(x) = \frac{x(\cos^{-1} x)^2}{2} + Cx$$

$$\lim_{x \rightarrow 1} \left[\frac{x(\cos^{-1} x)^2}{2} + Cx \right] = 1$$

$$\frac{1(0)^2}{2} + C(1) = 1 \Rightarrow C = 1$$

$$y(x) = \frac{x(\cos^{-1} x)^2}{2} + x$$

$$y\left(\frac{1}{2}\right) = \frac{\frac{1}{2} \left(\frac{\pi}{3}\right)^2}{2} + \frac{1}{2}$$

$$y\left(\frac{1}{2}\right) = \frac{\pi^2}{4 \cdot 9} + \frac{1}{2} = \frac{\pi^2}{36} + \frac{1}{2}$$

$$y\left(\frac{1}{2}\right) = \frac{\pi^2 + 18}{36}$$

Question: The value of $\int_0^2 \frac{\sqrt{x(x^2+x+1)}}{\sqrt{x+1}\sqrt{x^4+x^2+1}} dx$

Options:

(a) $\frac{1}{3} \ln(2^{3/2} + 3)$

(b) $\ln(2^{3/2} + 3)$

(c) $\ln(2^{3/2} + 1)$

(d) $\frac{2}{3} \ln(2^{3/2} + 3)$

Answer: (d)

Solution:

$$I = \int_0^2 \frac{\sqrt{x(x^2+x+1)}}{\sqrt{x+1}\sqrt{x^2+x+1}\sqrt{x^2-x+1}} dx$$

$$I = \int_0^2 \frac{\sqrt{x}\sqrt{x^2+x+1}}{\sqrt{x+1}\sqrt{x^2-x+1}} dx$$

$$I = \int_0^2 \sqrt{\frac{x(x^2+x+1)}{(x+1)(x^2-x+1)}} dx = \int_0^2 \sqrt{\frac{x^3+x^2+x}{x^3+1}} dx$$

$$I = \int_0^2 \sqrt{\frac{x+1+1/x}{x+1/x^2}} dx$$

$$I = \int_0^2 \sqrt{\frac{x(x^2+x+1)}{(x+1)(x^2-x+1)}} dx = \int_0^2 \sqrt{\frac{x^3+x^2+x}{x^3+1}} dx$$

$$I = \int_0^2 \sqrt{\frac{x+1+1/x}{x+1/x^2}} dx$$

Let $x = t^2$, then $dx = 2t dt$.

$$I = \int_0^{\sqrt{2}} \frac{t(t^4 + t^2 + 1)}{\sqrt{t^2 + 1} \sqrt{t^8 + t^4 + 1}} \cdot 2t dt$$

$$I = \ln(x^{3/2} + \sqrt{x^3 + 1}) \Big|_0^2$$

$$I = \ln(2^{3/2} + \sqrt{2^3 + 1}) - \ln(0 + \sqrt{1})$$

$$I = \ln(2^{3/2} + \sqrt{9}) - \ln(1)$$

$$I = \ln(2^{3/2} + 3)$$

Question: Let $\frac{x^2}{f(x^2 + 2a + 7)} + \frac{y^2}{f(3a + 14)} = 1$ represents an equation of ellipse. The major axis of given ellipse is y-axis and f is a decreasing function. If the range of a is $R - [\alpha, \beta]$ then $\alpha + \beta$ is

Options:

- (a) 3
- (b) 4
- (c) 2
- (d) 1

Answer: (d)

Solution:

$$\frac{x^2}{f(a^2 + 2a + 7)} + \frac{y^2}{f(3a + 14)} = 1$$

$$f(a^2 + 2a + 7) > 0$$

$$f(3a + 14) > 0$$

$$f(3a + 14) > f(a^2 + 2a + 7)$$

Since f is a **decreasing function**, the inequality $f(x_1) > f(x_2)$ implies $x_1 < x_2$.

$$3a + 14 < a^2 + 2a + 7$$

$$a^2 - a - 7 > 0$$

$$a = \frac{1 \pm \sqrt{(-1)^2 - 4(1)(-7)}}{2(1)} = \frac{1 \pm \sqrt{29}}{2}$$

The solution to the inequality $a^2 - a - 7 > 0$ is the region outside the roots:

$$a \in \left(-\infty, \frac{1 - \sqrt{29}}{2}\right) \cup \left(\frac{1 + \sqrt{29}}{2}, \infty\right)$$

In the notation $R - [\alpha, \beta]$, this corresponds to:

$$\alpha = \frac{1 - \sqrt{29}}{2} \quad \text{and} \quad \beta = \frac{1 + \sqrt{29}}{2}$$

$$\alpha + \beta = \frac{1 - \sqrt{29}}{2} + \frac{1 + \sqrt{29}}{2} = \frac{2}{2} = 1$$

Question: The number of values of $Z \in \mathbb{C}$ satisfying the equation $|z - (4 + i)| = \sqrt{10}$ and $|z - (3 + 5i)| + |z - (5 + 11i)| = 4\sqrt{5}$ is

Options:

- (a) 0
- (b) 1
- (c) 2
- (d) 4

Answer: (b)

Solution:

The equation $|z - (4 + i)| = \sqrt{10}$ represents a circle.

- Center $C: (4, 1)$

- Radius $R: \sqrt{10}$

The equation $|z - (3 + 5i)| + |z - (5 + 11i)| = 4\sqrt{5}$ represents the locus of points whose sum of distances from two fixed points is constant.

- Fixed point $A: (3, 5)$

- Fixed point $B: (5, 11)$

- Constant sum $2a: 4\sqrt{5}$

$$AB = \sqrt{(5-3)^2 + (11-5)^2} = \sqrt{2^2 + 6^2} = \sqrt{4 + 36} = \sqrt{40} = 2\sqrt{10} \approx 2 \times 3.16 = 6.32$$

$$\text{Sum} = 4\sqrt{5} \approx 4 \times 2.23 = 8.92$$

Since the constant sum is greater than the distance between the points ($4\sqrt{5} > 2\sqrt{10}$), the locus is an ellipse with foci A and B .

To find how many values of z satisfy both, we check if the center of the circle $C(4, 1)$ lies on the ellipse.

The sum of distances from C to the foci A and B is:

- $|C - A| = \sqrt{(4-3)^2 + (1-5)^2} = \sqrt{1^2 + (-4)^2} = \sqrt{17}$

- $|C - B| = \sqrt{(4-5)^2 + (1-11)^2} = \sqrt{(-1)^2 + (-10)^2} = \sqrt{101}$

- $\text{Sum} \approx 4.12 + 10.05 = 14.17$, which is much larger than $4\sqrt{5} \approx 8.92$.

Since the circle's radius is relatively small ($\sqrt{10} \approx 3.16$) and its center is far from the ellipse, we look for points of intersection. By checking the geometry or substituting specific values, it's found that the circle and ellipse intersect at exactly one point.

The correct answer is B) 1.

Question: $\vec{a} = 4\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{b} = 10\hat{i} + 2\hat{j} - \hat{k}$ and a vector $\vec{c}$ be such that $2(\vec{a} \times \vec{b}) + 3(\vec{b} \times \vec{c}) = \vec{0}$, if $\vec{a} \cdot \vec{c} = 15$, Then the value of $\vec{c} \cdot (\hat{i} + \hat{j} - 3\hat{k})$ is

Options:

- (a) 5
- (b) -5
- (c) -3
- (d) 3

Answer: (b)

Solution:

$$2(\vec{a} \times \vec{b}) + 3(\vec{b} \times \vec{c}) = \vec{0}$$

$$3(\vec{b} \times \vec{c}) = -2(\vec{a} \times \vec{b})$$

$$3(\vec{b} \times \vec{c}) = 2(\vec{b} \times \vec{a})$$

$$\vec{b} \times (3\vec{c} - 2\vec{a}) = \vec{0}$$

$$3\vec{c} - 2\vec{a} = \lambda \vec{b}$$

$$3\vec{c} = 2\vec{a} + \lambda \vec{b}$$

$$3(\vec{a} \cdot \vec{c}) = 2(\vec{a} \cdot \vec{a}) + \lambda(\vec{a} \cdot \vec{b})$$

$$\bullet \vec{a} = 4\hat{i} - \hat{j} + 3\hat{k} \implies \vec{a} \cdot \vec{a} = 4^2 + (-1)^2 + 3^2 = 16 + 1 + 9 = 26$$

$$\bullet \vec{b} = 10\hat{i} + 2\hat{j} - \hat{k} \implies \vec{a} \cdot \vec{b} = (4)(10) + (-1)(2) + (3)(-1) = 40 - 2 - 3 = 35$$

$$\bullet \vec{a} \cdot \vec{c} = 15$$

$$3(15) = 2(26) + \lambda(35)$$

$$\implies \lambda = -\frac{1}{5}$$

Question: The mean and variance of x_1, x_2, x_3, x_4 be 1 and 13 respectively and the mean and variance of $y_1, y_2, y_3, \dots, y_5, y_6$ be 2 and 1 respectively, then the variance of $x_1, x_2, x_3, x_4, y_1, y_2, \dots, y_6$ will be

Options:

(a) 6.04

(b) 6.58

(c) 5.96

(d) 6.25

Answer: (a)

Solution:

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$$

• n_1, n_2 : Number of observations in each set.

• σ_1^2, σ_2^2 : Variances of each set.

• d_1, d_2 : The difference between the individual group mean and the combined mean ($d_1 = \bar{x}_1 - \bar{x}$ and $d_2 = \bar{x}_2 - \bar{x}$).

• **Set 1 (x):** $n_1 = 4, \bar{x}_1 = 1, \sigma_1^2 = 13$

• **Set 2 (y):** $n_2 = 6, \bar{x}_2 = 2, \sigma_2^2 = 1$

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} = \frac{4(1) + 6(2)}{4 + 6} = \frac{4 + 12}{10} = \frac{16}{10} = 1.6$$

$$\bullet d_1 = 1 - 1.6 = -0.6 \implies d_1^2 = 0.36$$

$$\bullet d_2 = 2 - 1.6 = 0.4 \implies d_2^2 = 0.16$$

$$\sigma^2 = \frac{4(13 + 0.36) + 6(1 + 0.16)}{10}$$

$$\sigma^2 = \frac{4(13.36) + 6(1.16)}{10}$$

$$\sigma^2 = \frac{53.44 + 6.96}{10}$$

$$\sigma^2 = \frac{60.4}{10} = 6.04$$

Question: Let O be the vertex of the parabola $y^2 = 4ax$, ($a > 0$). Let P and Q be two variable points on the parabola such that chords OP and OQ are perpendicular to each other. If the locus of mid point of segment PQ is a coin C then latus rectum of C is

Options:

- (a) a
- (b) 2a
- (c) 3a
- (d) 4a

Answer: (b)

Solution:

Given the parabola $y^2 = 4ax$, let the vertex be $O(0, 0)$.

Let points P and Q on the parabola be:

- $P = (at_1^2, 2at_1)$
- $Q = (at_2^2, 2at_2)$

The chords OP and OQ are perpendicular. The slopes are:

- $m_{OP} = \frac{2at_1}{at_1^2} = \frac{2}{t_1}$
- $m_{OQ} = \frac{2at_2}{at_2^2} = \frac{2}{t_2}$

Since $OP \perp OQ$, the product of their slopes is -1 :

$$\left(\frac{2}{t_1}\right) \left(\frac{2}{t_2}\right) = -1 \implies t_1 t_2 = -4$$

Let the midpoint of PQ be (h, k) .

- $h = \frac{at_1^2 + at_2^2}{2} \implies 2h = a(t_1^2 + t_2^2)$
- $k = \frac{2at_1 + 2at_2}{2} \implies k = a(t_1 + t_2)$

We use the identity $(t_1 + t_2)^2 = t_1^2 + t_2^2 + 2t_1 t_2$:

- Substitute $t_1 + t_2 = \frac{k}{a}$ and $t_1^2 + t_2^2 = \frac{2h}{a}$
- Substitute $t_1 t_2 = -4$

$$\left(\frac{k}{a}\right)^2 = \frac{2h}{a} + 2(-4)$$

$$\frac{k^2}{a^2} = \frac{2h}{a} - 8$$

$$k^2 = 2ah - 8a^2$$

$$k^2 = 2a(h - 4a)$$

Replacing (h, k) with (x, y) , the locus equation is:

$$y^2 = 2a(x - 4a)$$

The equation $y^2 = 2a(x - 4a)$ is a parabola in the form $Y^2 = 4AX$, where the coefficient of the linear term (the latus rectum) is $2a$.

Question: A student goes to examination center by bus, scooter or car probability of which being equally likely. Probability that he reaches late, if he takes bus, scooter or car is $\frac{1}{5}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. Given that he reaches late, the probability he travelled by a bus is

Options:

- (a) $\frac{12}{47}$
- (b) $\frac{1}{2}$
- (c) $\frac{26}{7}$
- (d) $\frac{11}{6}$

Answer: (a)

Solution:

- Let B, S, C be the events that the student travels by **Bus, Scooter, and Car**, respectively.
 - Let L be the event that the student **reaches late**.
 - The modes of transport are equally likely:

$$P(B) = P(S) = P(C) = \frac{1}{3}$$
 - The conditional probabilities of being late are:

$$P(L|B) = \frac{1}{5}$$

$$P(L|S) = \frac{1}{3}$$

$$P(L|C) = \frac{1}{4}$$
- $$P(B|L) = \frac{P(B) \cdot P(L|B)}{P(B) \cdot P(L|B) + P(S) \cdot P(L|S) + P(C) \cdot P(L|C)}$$

$$P(B|L) = \frac{\frac{1}{3} \cdot \frac{1}{5}}{\left(\frac{1}{3} \cdot \frac{1}{5}\right) + \left(\frac{1}{3} \cdot \frac{1}{3}\right) + \left(\frac{1}{3} \cdot \frac{1}{4}\right)}$$

$$P(B|L) = \frac{\frac{1}{5}}{\frac{1}{5} + \frac{1}{3} + \frac{1}{4}}$$

$$P(B|L) = \frac{1/5}{47/60} = \frac{1}{5} \times \frac{60}{47} = \frac{12}{47}$$

Question: The line passing through points of intersection of $3x + 4y = 1$ and $4x + 3y = 1$ intersects axes at P and Q. Then the locus of mid point of PQ is

Options:

- (a) $\frac{1}{x} + \frac{1}{y} = 14$
- (b) $\frac{3}{x} + \frac{4}{y} = 14$

$$\frac{4}{x} + \frac{3}{y} = 14$$

- (c) $x + y = 14$

Answer: (a)

Solution:

Solve the given equations:

1. $3x + 4y = 1$

2. $4x + 3y = 1$

Subtracting (1) from (2): $(4x - 3x) + (3y - 4y) = 1 - 1 \Rightarrow x - y = 0 \Rightarrow x = y$.

Substituting $x = y$ into (1): $3x + 4x = 1 \Rightarrow 7x = 1 \Rightarrow x = 1/7$.

The point of intersection is **A(1/7, 1/7)**.

A line passing through **A(1/7, 1/7)** can be written in intercept form:

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\frac{1/7}{a} + \frac{1/7}{b} = 1 \Rightarrow \frac{1}{a} + \frac{1}{b} = 7 \quad \dots(\text{Eq. I})$$

- **P(a, 0)** on the x-axis.
- **Q(0, b)** on the y-axis.
- $x = \frac{a+0}{2} \Rightarrow a = 2x$
- $y = \frac{0+b}{2} \Rightarrow b = 2y$

$$\frac{1}{2x} + \frac{1}{2y} = 7$$

$$\frac{1}{x} + \frac{1}{y} = 14$$

Question: If the system of equation

$$x + y + z = 6$$

$$x + 2y + 5z = 10$$

$$x + 3y + \lambda z = \mu$$

Has infinitely many solutions then $\lambda + \mu$ is equal to

Answer: (23)

Solution:

Write the augmented matrix $[A|B]$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 5 & 10 \\ 1 & 3 & \lambda & \mu \end{array} \right]$$

$R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 4 & 4 \\ 0 & 2 & \lambda - 1 & \mu - 6 \end{array} \right]$$

$R_3 \rightarrow R_3 - 2R_2$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 4 & 4 \\ 0 & 0 & \lambda - 9 & \mu - 14 \end{array} \right]$$

For the system to have **infinitely many solutions**, the rank of the coefficient matrix must equal the rank of the augmented matrix, and both must be less than the number of variables (3). This occurs when the entire last row is zero:

- $\lambda - 9 = 0 \Rightarrow \lambda = 9$
- $\mu - 14 = 0 \Rightarrow \mu = 14$

The value of $\lambda + \mu$ is:

$$\lambda + \mu = 9 + 14 = 23$$

Question: Let $f(x) = \begin{cases} \frac{1}{3} & , x < \frac{\pi}{2} \\ \frac{b(1-\sin x)}{(\pi-2x)^2} & x > \frac{\pi}{2} \end{cases}$ if f is a continuous function at $x = \frac{\pi}{2}$, then

the value of $\int_0^{3b-6} |x^2 + 2x - 3| dx$ is

Answer: (4)

Solution:

For $f(x)$ to be continuous at $x = \frac{\pi}{2}$, the left-hand limit (LHL) must equal the right-hand limit (RHL).

- LHL ($x \rightarrow \frac{\pi}{2}^-$): $f(x) = \frac{1}{3}$.
- RHL ($x \rightarrow \frac{\pi}{2}^+$): $\lim_{x \rightarrow \frac{\pi}{2}^+} \frac{b(1-\sin x)}{(\pi-2x)^2}$

Let $x = \frac{\pi}{2} + h$, where $h \rightarrow 0$:

$$\lim_{h \rightarrow 0} \frac{b(1-\sin(\frac{\pi}{2} + h))}{(\pi-2(\frac{\pi}{2} + h))^2} = \lim_{h \rightarrow 0} \frac{b(1-\cos h)}{(-2h)^2}$$

$$\lim_{h \rightarrow 0} \frac{b \cdot \frac{h^2}{2}}{4h^2} = \frac{b}{8} \quad \text{Using the identity } 1 - \cos h \approx \frac{h^2}{2} \text{ for small } h:$$

Equating LHL and RHL:

$$\frac{1}{3} = \frac{b}{8} \Rightarrow b = \frac{8}{3}$$

Substitute $b = \frac{8}{3}$ into the upper limit: $3b - 6 = 3(\frac{8}{3}) - 6 = 8 - 6 = 2$.

The integral is:

$$I = \int_0^2 |x^2 + 2x - 3| dx$$

$$I = \int_0^1 -(x^2 + 2x - 3) dx + \int_1^2 (x^2 + 2x - 3) dx$$

$$I = \left[-\frac{x^3}{3} - x^2 + 3x \right]_0^1 + \left[\frac{x^3}{3} + x^2 - 3x \right]_1^2$$

$$I = \left(-\frac{1}{3} - 1 + 3 \right) + \left(\frac{8}{3} + 4 - 6 \right) - \left(\frac{1}{3} + 1 - 3 \right)$$

$$I = \frac{5}{3} + \left(\frac{2}{3} - \left(-\frac{5}{3} \right) \right) = \frac{5}{3} + \frac{7}{3} = \frac{12}{3} = 4$$

Question: Consider the relation R defined on the set $A = \{-2, -1, 0, 1, 2\}$ defined by $(a, b) \in R$ if and only $1 + ab > 0$.

Given below are two statement:

Statement I : The number of elements in R is 17.

Statement II : R is an equivalence relation in light of the above statements, choose the correct answer.

Options:

- (a) Both Statement I and Statement II are correct
- (b) Both Statement I and Statement II are incorrect
- (c) Statement I is correct but Statement II is incorrect
- (d) Statement I is incorrect but Statement II is correct

Answer: (c)

Solution:

To determine which option is correct, let's analyze both statements step-by-step based on the set $A = \{-2, -1, 0, 1, 2\}$ and the relation $R = \{(a, b) : 1 + ab > 0\}$.

Statement I: The number of elements in R is 17

We need to find all pairs (a, b) such that $ab > -1$.

- For $a = -2$: b can be $-2, -1, 0$ (since $-2 \cdot -2 = 4, -2 \cdot -1 = 2, -2 \cdot 0 = 0$). (3 pairs)
- For $a = -1$: b can be $-2, -1, 0$ (since $-1 \cdot -2 = 2, -1 \cdot -1 = 1, -1 \cdot 0 = 0$). (3 pairs)
- For $a = 0$: b can be any element in A (since $0 \cdot b = 0$, and $1 + 0 > 0$). (5 pairs)
- For $a = 1$: b can be $0, 1, 2$ (since $1 \cdot 0 = 0, 1 \cdot 1 = 1, 1 \cdot 2 = 2$). (3 pairs)
- For $a = 2$: b can be $0, 1, 2$ (since $2 \cdot 0 = 0, 2 \cdot 1 = 2, 2 \cdot 2 = 4$). (3 pairs)

Total number of elements = $3 + 3 + 5 + 3 + 3 = 17$.

Thus, **Statement I is correct.**

Statement II: R is an equivalence relation

An equivalence relation must be reflexive, symmetric, and transitive.

1. **Reflexive:** $1 + a^2 > 0$ is true for all $a \in A$. (Checked)
2. **Symmetric:** If $1 + ab > 0$, then $1 + ba > 0$ is also true because $ab = ba$. (Checked)
3. **Transitive:** If $(a, b) \in R$ and $(b, c) \in R$, then (a, c) must be in R .
 1. Let's test $a = -2$, $b = 0$, and $c = 2$.
 2. $(-2, 0) \in R$ because $1 + (-2)(0) = 1 > 0$.
 3. $(0, 2) \in R$ because $1 + (0)(2) = 1 > 0$.
 4. However, $(-2, 2) \notin R$ because $1 + (-2)(2) = -3$, which is **not** greater than 0.

Since it is not transitive, **Statement II is incorrect.**

Statement I is correct but Statement II is incorrect.

The correct option is **C**.

